

Logos Experiment 15 Report

Large Scaling with Weight Tying + SwiGLU, trained for 5000 iterations

1. Experiment Objective

Experiment 15 extends the large Logos model from Experiment 14 by increasing the training duration from 3000 to 5000 iterations. The purpose is to test whether the 124M-parameter model was undertrained and whether additional optimization can improve validation loss and perplexity without changing the architecture, parameter count, checkpoint size, or VRAM footprint.

2. Model and Training Setup

Component	Value
Run name	logos_exp15_large_scaling_weight_tying_swiglu_longer_train
Model family	Decoder-only GPT-style causal language model
Dataset	OpenWebText subset, 100,000 samples
Tokenizer	GPT-2 tokenizer
Architecture recipe	Weight Tying + SwiGLU
Layers	12
Embedding dimension	768
Attention heads	12
Head dimension	64
Context length	512 tokens
Batch size	32
Tokens per step	16,384
Training iterations	5000
Optimizer	AdamW
Learning rate	3e-4
Warmup iterations	250

3. Final Results

Metric	Experiment 15 Result
Parameters	124,031,232
Best iteration	5000
Best validation loss	4.4690
Final validation loss	4.4663
Final validation perplexity	87.03
Checkpoint size	473.20 MB
Peak VRAM	17.73 GB
Training time	1187.97 sec = 19.80 min
Generation speed	59.67 tokens/sec
Generation latency	16.76 ms/token

Metric	Experiment 15 Result
Average GPU power	291.33 W
Estimated GPU energy	96.03 Wh

4. Comparison with Experiment 14

Metric	Exp 14: 3000 iters	Exp 15: 5000 iters	Outcome
Parameters	124.03M	124.03M	unchanged
Dataset	100k	100k	unchanged
Context	512	512	unchanged
Best val loss	4.9201	4.4690	improved
Final val loss	4.9244	4.4663	improved
Final PPL	137.60	87.03	improved
Checkpoint	473.20 MB	473.20 MB	unchanged
Peak VRAM	17.73 GB	17.73 GB	unchanged
Training time	12.43 min	19.80 min	higher
Energy	59.76 Wh	96.03 Wh	higher

5. Scientific Interpretation

Experiment 15 is a major positive result. The same 124.03M-parameter model achieved a much lower validation loss and perplexity after longer training, showing that the Experiment 14 model was clearly undertrained at 3000 iterations. The improvement did not require additional parameters, a larger checkpoint, a longer context window, or more VRAM.

The main tradeoff is expected: longer training increased wall-clock time and energy consumption. However, from a capability-per-parameter and capability-per-checkpoint perspective, Experiment 15 is the strongest model so far.

6. Key Conclusion

Experiment 15 is the current best overall quality model in the Logos project. It achieved a final validation perplexity of 87.03 with 124.03M parameters and a 473.20 MB checkpoint, while using only 17.73 GB of peak VRAM on the RTX 6000 Ada system.

7. Next Recommended Step

The next scientific step is to update the master leaderboard and then run a controlled context-length experiment. Since Experiment 15 still used context length 512, the next useful test is whether the same 124M model benefits from a 1024-token context window while keeping the training recipe consistent.